Customer Churn Prediction Using Elastic Net Regularized Logistic Regression

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Section9

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1. Problem Definition

Customer Churn Prediction

- Customer churn occurs when a customer stops using or subscribing to a company's service.
- High customer churn can lead to revenue loss and increased customer acquisition costs.
- It is difficult for organizations to identify customers who are likely to leave in advance.
- Machine learning can analyze historical customer data and identify customers who have a higher probability of churning.
- This project develops a Customer Churn Prediction system using Elastic Net Regularized Logistic Regression.

Why is this problem important?

- Helps organizations identify high-risk customers.
- Supports customer retention strategies.
- Reduces potential revenue loss.
- Helps businesses understand factors associated with churn.
- Enables data-driven decision-making.

Who will use the solution?

- Telecommunications companies
- Customer relationship management teams
- Marketing and sales teams
- Business analysts
- Customer retention teams

Expected Output

- The system will:
- Input: Customer information
- ↓
- Machine Learning Model
- ↓
- Output:
- Churn Probability
- Churn / No Churn prediction
- Identification of high-risk customers

2. Dataset Collection

Dataset Information

- **Dataset Name:** IBM Telco Customer Churn Dataset
- **Dataset**
  - **File:** WA_Fn-UseC_-Telco-Customer-Churn.csv
- **Dataset Source:** IBM sample/customer churn dataset, commonly distributed through Kaggle and other public dataset repositories.
- **Number of Records:** 7,043
- **Number of Columns:** 21
- **Target Variable:** Churn
- **Target Values:**
- Yes → Customer churned
- No → Customer did not churn

Dataset Features:

Category	Features
Customer Information	customerID, gender, SeniorCitizen, Partner, Dependents
Service Information	PhoneService, MultipleLines, InternetService
Additional Services	OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport
Streaming Services	StreamingTV, StreamingMovies
Contract & Payment	Contract, PaperlessBilling, PaymentMethod
Financial	MonthlyCharges, TotalCharges
Target	Churn

3. Data Understanding

Basic Dataset Analysis

- From the uploaded dataset:
- **Records:** 7,043
- **Features:** 20 input features
- **Target:** 1 (Churn)
- **Total columns:** 21
- **Duplicate records:** 0
- **Missing values:** No direct NaN values detected

Numerical Variables	Categorical Variables
SeniorCitizen tenure MonthlyCharges	gender Partner Dependents PhoneService MultipleLines InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV StreamingMovies Contract PaperlessBilling PaymentMethod Churn

Statistical Summary

- For numerical features, statistical analysis can include:
- Mean
- Median
- Standard deviation
- Minimum
- Maximum
- Quartiles
- Important numerical variables:

tenure

- Represents the number of months a customer has stayed with the company.

MonthlyCharges

- Represents the customer's monthly service charge.

TotalCharges

- Represents the total amount charged to the customer.

Purpose: Statistical summary helps understand the range, central tendency, and variation of numerical variables.

Unique Values

Gender

- Male
- Female

Contract

- Month-to-month
- One year
- Two year

Payment Method

- Mailed check
- Bank transfer
- Credit card

Churn

- Yes
- No
- This helps us understand the categories available in the dataset.

4. Data Preprocessing

1. Remove irrelevant attributes

- Remove:
- customerID
- because it is only a unique customer identifier.

2. Handle missing/blank values

- Check for missing and blank values.
- For example:
- TotalCharges
- is initially stored as text and needs conversion to numerical format.

3. Convert numerical variables

- Convert:
- TotalCharges
- from object/string to numeric.

4. Handle categorical variables

- Categorical variables cannot be directly given to Logistic Regression.
- Therefore, encoding techniques such as **One-Hot Encoding** are used.
- Example:
- Contract
- Month-to-month
- One year
- Two year
- becomes numerical indicator columns.

5. Scale numerical features

- Numerical features such as:
- tenure
- MonthlyCharges
- TotalCharges
- are scaled so that variables with different ranges can be handled effectively by the model.

```
missing_values = df.isnull().sum()

print("Missing values in each column:")
print(missing_values)
```

```
Missing values in each column:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

```
# Convert TotalCharges to numeric
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')

# Check the data type
print(df['TotalCharges'].dtype)

# Check missing values created during conversion
print("Missing TotalCharges:", df['TotalCharges'].isnull().sum())
```

```
float64
Missing TotalCharges: 11
```

+ Code + Text

```
#checking outliers
```

```
categorical_columns = df.select_dtypes(include=['object']).columns

print("Categorical Variables:")
print(categorical_columns.tolist())
```

```
Categorical Variables:
['customerID', 'gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBack
```

```
numerical_columns = df.select_dtypes(include=['int64', 'float64']).columns

print("Numerical Variables:")
print(numerical_columns.tolist())
```

```
Numerical Variables:
['SeniorCitizen', 'tenure', 'MonthlyCharges']
```

Outlier and Incorrect Data Checking

Outlier Detection

- Box plots can be used to identify unusual values in:
- tenure , MonthlyCharges , TotalCharges

Incorrect Data

Check for:

- Blank values
- Invalid numerical values
- Unexpected categorical values
- Incorrect data types

Final preprocessing flow

Raw Dataset

↓

Remove customerID

↓

Handle blank/missing values

↓

Convert numerical columns

↓

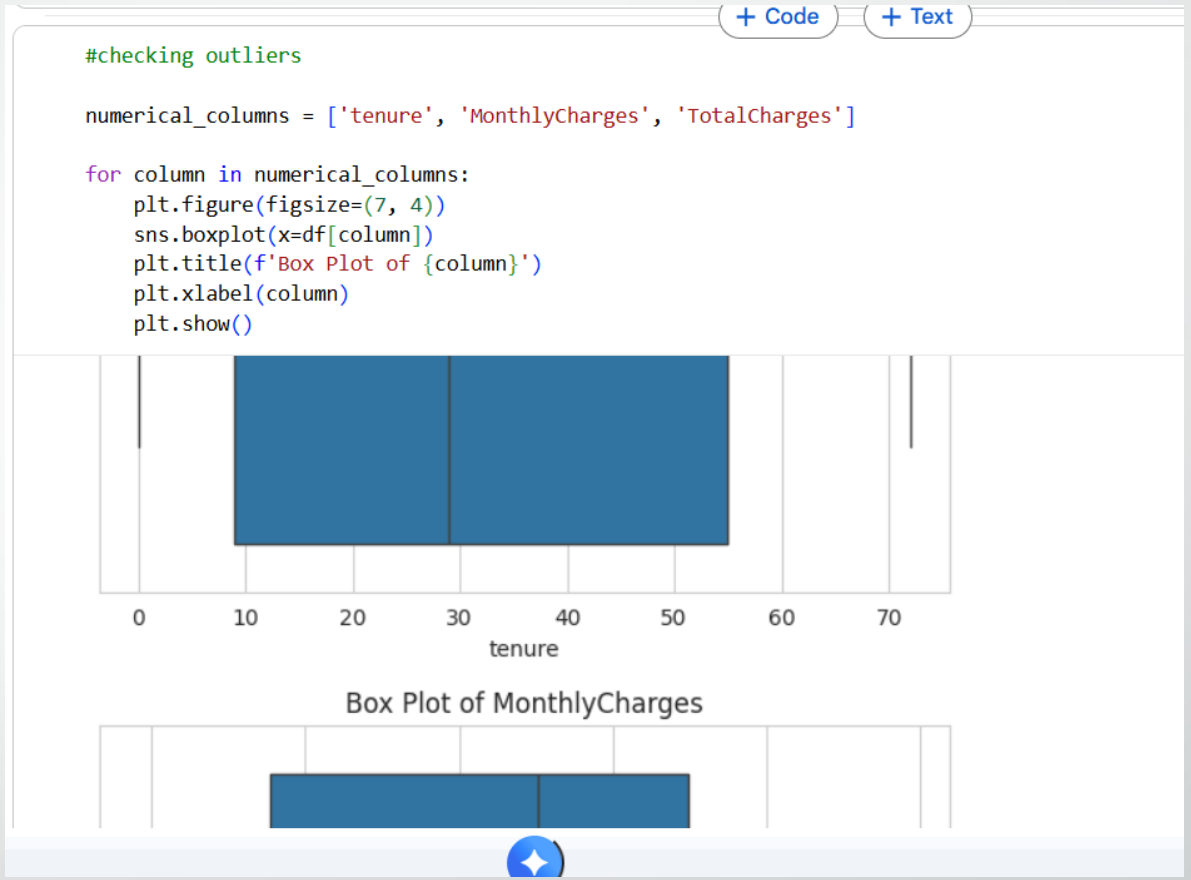
Encode categorical variables

↓

Scale numerical variables

↓

Preprocessed Dataset



```
numerical_columns = ['tenure', 'MonthlyCharges', 'TotalCharges']

for column in numerical_columns:

    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)

    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR

    outliers = df[
        (df[column] < lower_bound) |
        (df[column] > upper_bound)
    ]

    print(f"{column}")
    print("Lower bound:", lower_bound)
    print("Upper bound:", upper_bound)
    print("Number of potential outliers:", len(outliers))
    print()
```

```
tenure
Lower bound: -60.0
Upper bound: 124.0
Number of potential outliers: 0

MonthlyCharges
Lower bound: -46.02499999999999
Upper bound: 171.375
Number of potential outliers: 0

TotalCharges
Lower bound: -4674.3375
Upper bound: 8863.1625
Number of potential outliers: 0
```

```
#outilers
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    Q1 = df[column].quantile(0.25)
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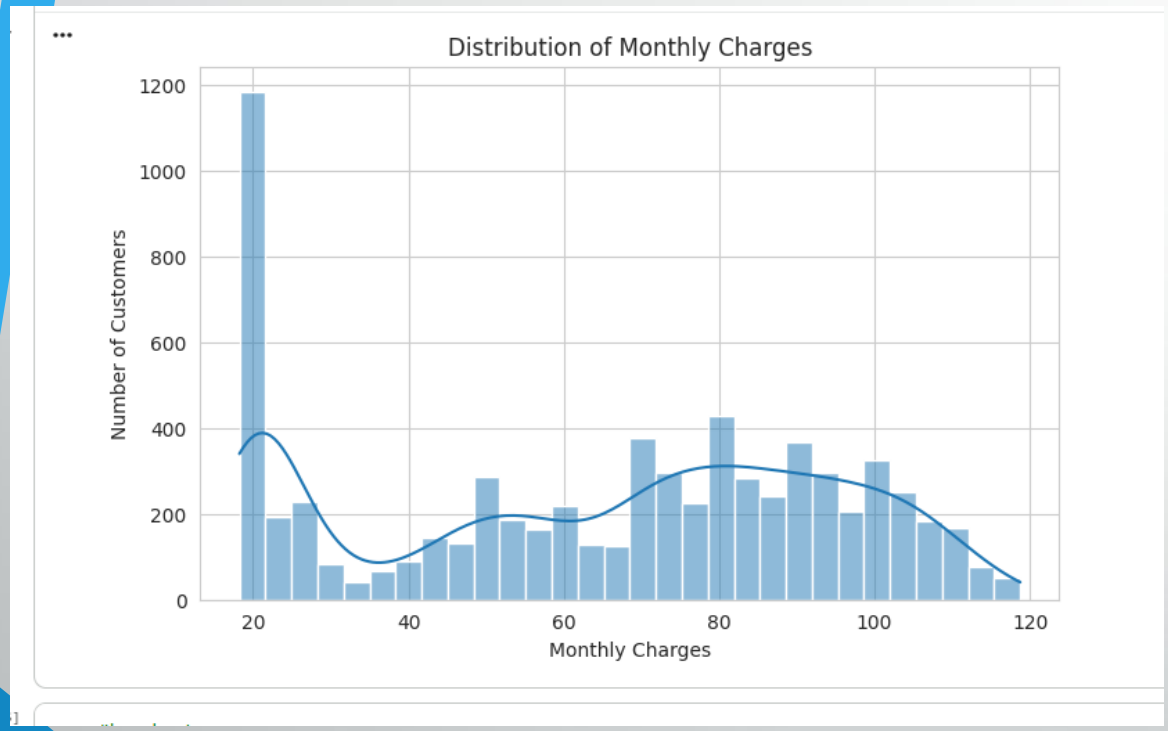
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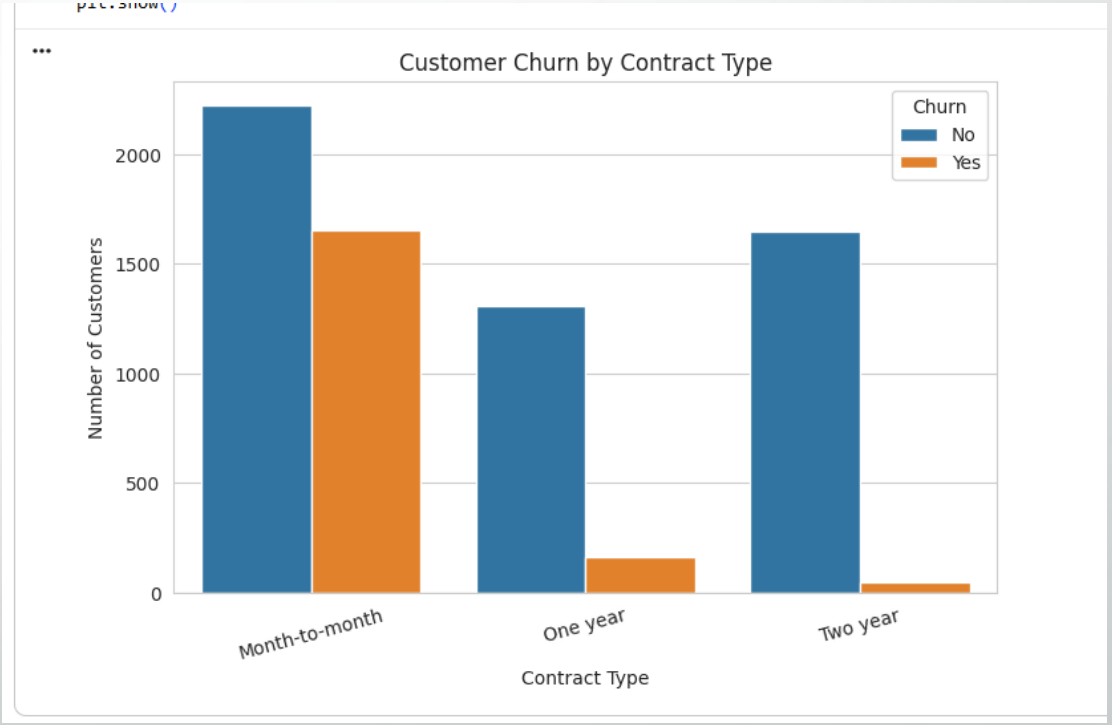
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Lower bound: -4674.3375
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Number of potential outliers: 0
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5. Exploratory Data Analysis — EDA

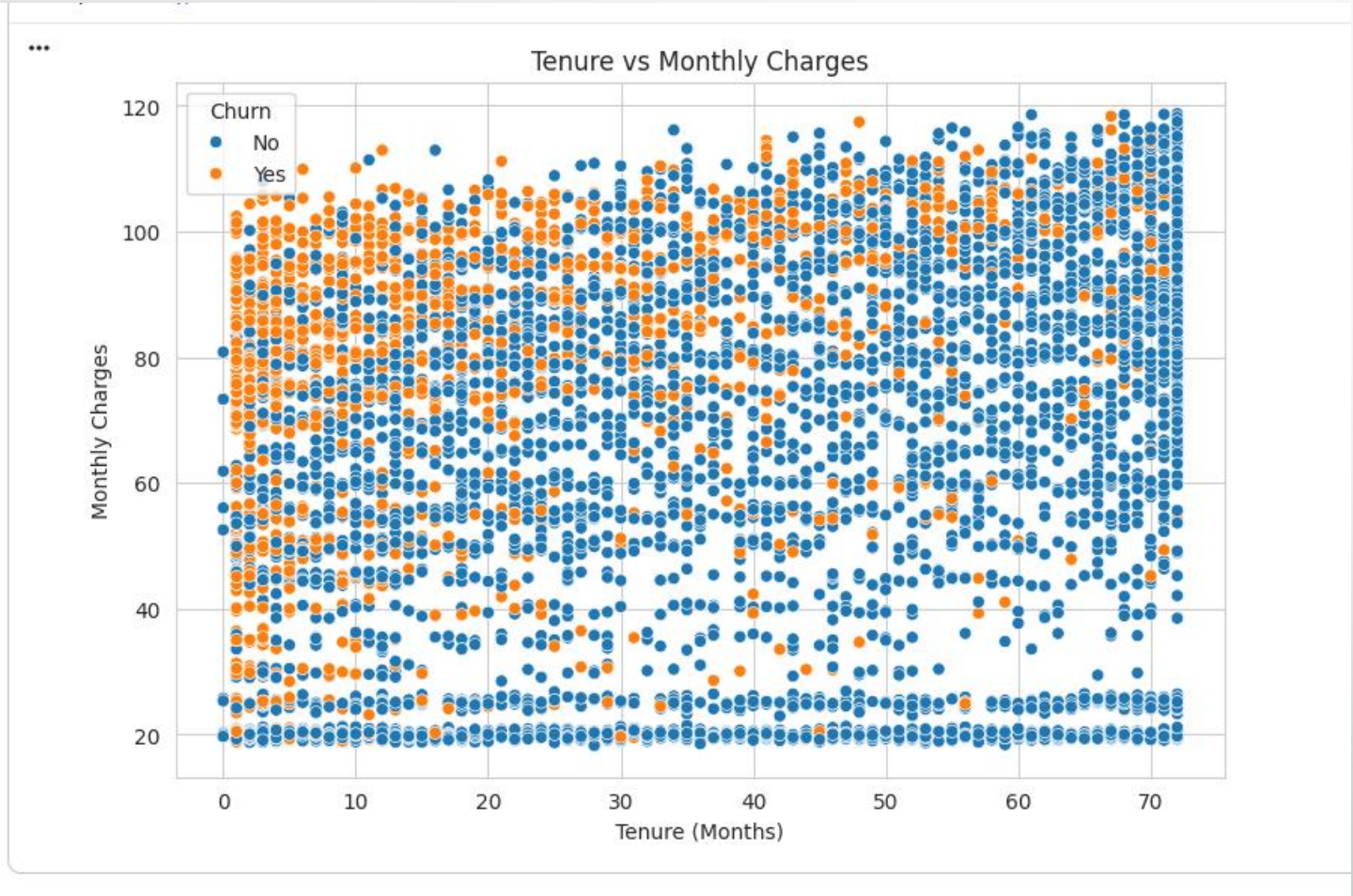
Histogram



Bar chart



Scatter plot



THANK YOU